

Digital Event Platform — Refined System Specification & Incremental Build Plan

This document defines a **compound, production-grade system** for managing digital invitations, RSVPs, check-in, and a digital guestbook across multiple events. It is written to provide **sufficient context for AI builders (e.g., Cursor, GPT-based tools)** while remaining durable, extensible, and aligned with a **premium, managed-service model**.

1. System Overview

The platform is a compound, admin-controlled event management system. Each event is modular: **admins explicitly choose which services are enabled per event**.

For every event, the admin can independently enable or disable the following services: - Invitation Website - RSVP System - Digital Guestbook

An event may use any combination of these services (e.g. RSVP-only, Guestbook-only, or full bundle).

All downstream behavior (pages exposed, QR codes generated, guest access, admin views) must respect the enabled services for that event.

The platform is an **internally operated event system** where admins create and manage multiple events. Each event acts as a container that combines:

- Custom-designed invitation websites
- Integrated RSVP flows (with approval gating if required)
- Secure on-site check-in
- Digital guestbook (audio, video, photos)
- Post-event thank-you pages and media access

This is **not** a self-serve SaaS. All configuration and setup is handled by internal admins. Couples/clients have **limited, read-only + approval visibility**.

2. Core Architectural Principle

Events reference templates. Templates render pages.

- Events store data, rules, and state
- Templates store presentation (HTML/CSS/JS)
- Pages are rendered by injecting event data into selected templates

This separation is mandatory for scalability and customization.

3. Event Model

Each event includes:

- Event metadata (name, date, time, type)
- Event phase (pre-event, live, post-event)
- Invitation-only flag
- RSVP rules
- Assigned templates per page type
- Feature flags (guestbook enabled, check-in enabled, etc.)

Event Phase Control

Event phase can be: - **Automatically set** based on event date and time - **Manually overridden** by admins at any time

The active phase strictly controls what guests can see or do.

4. Event Phases & Guest Capabilities

Phase	Guest Capabilities
Pre-event	View invitation website, submit RSVP
Live event	Record messages, upload photos
Post-event	View thank-you page only

Guests must never access features outside the active phase.

5. Template System (Critical)

Template Types

The system supports four template categories:

1. Invitation Website Template
2. RSVP Page Template
3. Guestbook Page Template
4. Thank-You Page Template

Template Characteristics

- Plain HTML / CSS / JS
- Uploaded and managed by admins
- Reusable across events
- Versionable and previewable
- Can be added or removed over time

Template Assignment

For each event, admins select which template to use for each page type.

Templates receive injected data: - Event metadata - CTA URLs (RSVP, Guestbook, Booth, Thank-you) - Phase-aware flags

Templates contain **no business logic**.

6. RSVP System

RSVP Submission

Guests submit: - Name(s) - Attendance (Yes / No / Maybe) - Guest count (optional) - Meal preference (optional) - Optional note

No authentication required.

Invitation-Only Events

If `invitation_only = true`:

- All RSVPs enter **Pending** state
- Couples/admins must approve or reject

RSVP lifecycle:

Submitted → Pending → Approved → Invitation Issued
 ↘ Rejected

7. RSVP Review & Approval (Couple Portal)

Couples access a secure, tokenized portal to:

- View RSVPs in near-real-time
- Filter by status (Pending / Approved / Rejected)
- Approve or reject RSVPs

They **cannot** edit or delete records.

8. Invitation Pass Issuance

Upon approval, the system generates:

- Unique QR code
- Unique 6-digit numeric access code
- Printable / downloadable invitation card (PDF)

Delivery Channels (Configurable)

- Email
- SMS (includes 6-digit code)
- WhatsApp

Invitation cards include: - Event name & date - Guest name(s) - QR code - 6-digit manual code

9. RSVP Rejection Handling

Timing

- Rejection message is sent **immediately** when an RSVP is rejected

Channel

- Same channel used for RSVP submission (email, SMS, WhatsApp)
- If no channel is available, show on-screen confirmation only

Message (Fixed for MVP)

“Thank you for your response. The event organizers will be in touch.”

Rejected guests receive no invitation credentials and cannot access check-in or guestbook features.

10. Check-In System

Access

- Web-based
- **Mobile-first**, then **tablet-optimized**

Verification Methods

1. QR code scanning (device camera or Bluetooth scanner)
2. Manual entry of 6-digit code

Validation Rules

- Code/QR must be valid, approved, and unused
 - Successful check-in marks attendee as checked in
 - Duplicate attempts are rejected
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11. Guestbook System (Live Phase Only)

Access Modes

- Personal device (QR)
- Booth mode (kiosk-style)

Message Flow

Guests choose: - **Video message** or - **Audio message**

System behavior: - When video is selected, automatically request and enable device camera - One-tap record button - Visible timer - Enforced duration limits (30–120 seconds) - Automatic upload after recording

No previews or editing tools in MVP.

Photo Uploads

- Guests can upload photos
 - Enforce per-device limits (plan-based)
 - Show remaining quota before upload
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12. Media Access (Couple Portal)

Couples can: - View media timeline - Play video/audio inline - View photo gallery - Download all media as ZIP - Download compiled video reel (if enabled)

No deletion or moderation controls.

13. Internal Admin System

Admins manage:

- Events (create, update, archive)
- Templates (upload, preview, assign, delete)
- Event phases (automatic + manual override)
- RSVP rules and approvals
- Messaging channels and limits
- Media storage and exports

This system is not exposed to clients.

14. Non-Functional Requirements

- Mobile-first UX
 - Extremely lightweight
 - Graceful handling of poor internet
 - Secure tokenization
 - Durable architecture
 - Easy customization via templates
 - Visually polished by design
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15. Incremental Build Plan (AI-Friendly)

Phase A — Core Foundation

- Admin authentication
- Event CRUD
- Event phase logic (auto + manual override)
- Template upload & management
- Template-to-event assignment
- Page rendering engine

Phase B — RSVP Engine

- RSVP endpoints
- Invitation-only approval flow
- Couple read-only portal (RSVP visibility)
- Rejection messaging
- Invitation credential generation

Phase C — Check-In System

- Mobile-first check-in UI

- QR scanning
- Manual code entry
- Attendance tracking

Phase D — Guestbook MVP

- Video/audio recording with camera auto-enable
- Photo uploads with limits
- Booth mode (kiosk behavior)
- Media storage

Phase E — Media & Exports

- Media timeline
- ZIP download
- Video reel generation
- UI polish and reliability hardening

Phase F — Messaging Add-On (Optional)

- Broadcast announcements
- Delivery logs
- Channel limits

Design Principle: When trade-offs exist, always prioritize simplicity, reliability, clarity, and premium user experience over feature depth.

INCREMENTAL BUILD PLAN (MODULAR, FEATURE-FLAG DRIVEN)

This system must be built incrementally, with **event-level modularity enforced from the very first step**.

No feature may assume another feature exists unless it is explicitly enabled for the event.

Increment 1: Core Platform Foundation

- Platform admin authentication
- Event creation and management
- Event metadata:
 - Name

- Date & time
- Timezone
- **Event service configuration (feature flags):**
 - Invitation Website (on/off)
 - RSVP System (on/off)
 - Digital Guestbook (on/off)
- Phase engine:
 - Phases: Pre-Event, Event Live, Post-Event
 - Automatic phase switching based on event date/time
 - Manual admin override

No public pages are exposed in this increment.

Increment 2: Template Management System

- Admin-managed template registry
- Templates are simple HTML/CSS/JS bundles
- Supported template types:
 - Invitation Page
 - RSVP Page
 - Thank-You Page
 - Guestbook Page
- Admin capabilities:
 - Upload templates
 - Delete templates
 - Preview templates
- Per event:
 - Select which template is used for each **enabled** service

Templates must be swappable without redeploying the system.

Increment 3: Invitation Website (If Enabled)

- Public event URL renders the assigned Invitation template
 - Dynamic injection of CTAs:
 - RSVP link (if RSVP enabled)
 - Guestbook link (if Guestbook enabled and allowed by phase)
 - Templates fully control design and layout
 - No hardcoded UI assumptions
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Increment 4: RSVP System (If Enabled)

- RSVP form embedded within assigned RSVP template
- RSVP lifecycle:
 - Submitted → Approved → Invitation Issued
 - Submitted → Rejected
- Admin RSVP review interface:
 - View pending RSVPs
 - Approve or Reject
- Invitation issuance on approval:
 - SMS with **6-digit unique code**
 - Email / WhatsApp with QR + link
 - Printable / downloadable invitation card
- Rejection handling:
 - Immediate on-screen message after submission:

“Thank you for your response. The event organizers will be in touch.”

- Optional follow-up message via same channel used for RSVP

Rejected RSVPs never receive a valid invitation token.

Increment 5: Check-In System (If RSVP Enabled)

- Mobile-first check-in interface
 - Tablet-optimized layout
 - Two verification modes:
 - QR code scan
 - Manual entry of 6-digit invitation code
 - Real-time validation against approved invitation tokens
 - Offline-tolerant UX where possible
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Increment 6: Digital Guestbook (If Enabled)

- Public access via QR or link
- Access enforced by:
 - Event phase
 - RSVP approval (for invitation-only events)
- Guest flow:
 - Optional name entry
 - Choose **Video** or **Audio**
 - Automatic camera/microphone prompt on selection
 - One-tap record
 - Visible timer
 - Hard duration limits
 - Immediate upload
 - Photo uploads with per-device limits
- **Booth Mode:**
 - Full-screen kiosk UI
 - No navigation escape
 - Auto-reset after submission

Increment 7: Admin Media Management

- Event dashboard media overview
- Timeline feed:
 - Guest name (if provided)
 - Media type
 - Timestamp
- Inline playback
- “Download All” (ZIP)
- Premium:
 - Asynchronous video reel generation
 - Admin notification when ready
 - Reel download

Increment 8: System Hardening

- Graceful handling of poor connectivity
- Upload retries / resumable uploads
- Backend enforcement of all limits
- Audit logs:
 - RSVP approvals
 - Rejections
 - Check-ins